DG INTERNS HUB — TECHNICAL NOTES

TOPIC 11: COMMON SIEM USE CASES IN SOC OPERATIONS

Submitted By: Gunjan
Internship: DG Interns Hub Cybersecurity
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1. Core SOC Use Cases Catalog

- **Privilege Escalation:** Monitoring unauthorized additions of accounts to Domain Admins or Shadow Administrators (Event ID 4728 / 4732).
- **Lateral Movement:** Detecting remote execution using PsExec, SMB pipe creation, or WinRM sessions across non-admin workstations.
- **Data Exfiltration:** Detecting large volume transfers over unauthorized protocols (DNS tunneling or SSH port forwarding).
- **Ransomware Indicators:** Sudden deletion of Volume Shadow Copies (`vssadmin delete shadows`) accompanied by massive file rename activity.

2. Practical SOC Scenario: Ransomware Outbreak Triage

Scenario: Detecting Shadow Copy Deletion and Mass File Encrypting

Rule Trigger: Sysmon Event ID 1 captures command execution: `vssadmin.exe Delete Shadows /All /Quiet`.

SIEM Correlation: Combines process launch alert with endpoint FIM generating 500 file modification alerts within 30 seconds.

Analyst Workflow: High severity P1 incident confirmed. Analyst immediately triggers SOAR endpoint isolation playbook to prevent lateral spread over SMB.

3. Conclusion

A mature SOC builds and tunes SIEM use cases aligned directly with the MITRE ATT&CK framework, ensuring effective coverage across initial access, execution, persistence, and exfiltration.